

The Geometry of Reduction

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Abstract

The document describes the structural relation between the full arrangement of what is and that slice of it which is accessible to observation. It formulates this relation in general terms, without attachment to any concrete physical, philosophical, or psychological content. The geometry of reduction describes four levels of structural relation: between S and $R(S)$, the division of components into symmetric and asymmetric, the parametric space of points of reduction with a subset of points allowing an observer, and the double nesting of reductions — the external $S \rightarrow R(S)$ and the internal one, structuring separate dimensions of $R(S)$ through the observer's memory.

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1. The nature of this material

This document describes the structural relation between the full arrangement of what is and its slice accessible to observation. It states the relation in general terms, without binding it to any particular physical, philosophical, or psychological content. Concrete applications are possible, but they are not examined here.

2. The full structure and the reduced description

The full structure S is the arrangement of what is, taken in its entire composition.

The observable description is $R(S)$: the reduced form of S , where R is the reduction operator that transforms the full structure into a form suitable for observation.

Reduction, by its meaning, is a transformation with loss. If there is no loss, the transformation is complete. It follows that a difference always holds between S and $R(S)$, and this difference is constitutive of the concept of reduction.

This text introduces no new entities. It fixes a structural relation that holds whenever the observable is separated from the full arrangement of the observable.

3. Symmetric and asymmetric components

Components of S that are symmetric relative to R yield stable forms under reduction. These forms reproduce identically at every application of R and constitute what in $R(S)$ reads as laws, regularities, invariants.

Components of S that are asymmetric relative to R yield residues under reduction — traces not reducible to stable forms. In $R(S)$ they read as anomalies, residual tensions, parameters not explainable by the laws derived.

The distinction between the symmetric and asymmetric parts of S is a distinction relative to R . One and the same component of S can be symmetric relative to one reduction and asymmetric relative to another.

Hence the composition of laws and the composition of anomalies in $R(S)$ is determined not only by S but also by the choice of R .

4. The space of points of reduction

The reduction operator has a parametrization: the set of possible ways to perform reduction from S . Each value of the parameter corresponds to its own point of reduction, and at each such point its own $R(S)$ arises.

The space of points of reduction is a parametric space. Points in it have the status of possibilities. A statement about this space is a statement about the structure of S , which admits different ways of reduction — not a statement about a multitude of separate worlds.

Each point of reduction has its own composition of symmetric and asymmetric components. It follows that both the composition of laws and the composition of anomalies in $R(S)$ differ from point to point.

The content of a particular point of reduction — what $R(S)$ looks like in it — is not derivable from the geometry.

5. The observer

$R(S)$ acquires determinacy as a describable structure only when an observer is present at the point.

The observer is that through which $R(S)$ becomes a structure with content. Without an observer, the point of reduction remains a point in parametric space but generates no describable slice of what is.

Not every point of reduction admits the formation of an observer. The subset of points at which an observer is possible is a proper subset within the space of all points of reduction.

This subset is not singled out by external selection. It follows from which points are compatible with the assembly of a structure capable of serving as observer.

6. Internal reduction

At a point that admits an observer, besides the external reduction $S \rightarrow R(S)$, an internal reduction operates: through it, individual dimensions of $R(S)$ acquire their structure.

Internal reduction performs the same function relative to a single dimension of $R(S)$ that external reduction performs relative to S as a whole: it transforms material into a structured form with loss.

The internal reduction is carried by the function of holding and continuity: memory. Memory does not create the dimension to which it applies. It structures that dimension.

It follows that at a point that admits an observer there are at least two nested reductions: the external one, which separates $R(S)$ from S , and the internal one, which forms structure inside $R(S)$ through the work of the observer's memory.

7. The structurable dimension

Time in the $R(S)$ of a point that admits an observer is a dimension that acquires structure through the work of internal reduction.

Time as material may be present in $R(S)$ as one of its components. The structure of time — sequence, directedness, the distinction between before and after — is not present in the material on its own. It arises through the observer's memory.

Therefore the claim that time has structure pertains to a point that admits an observer, not to S as such. At points that do not admit an observer, there is no ground for ascribing structure to time, and therefore no ground for speaking of temporal unfolding as such.

Without the structuring function, the category of time in its full sense does not apply to the point.

8. The limit of descriptive means

Entities may be present in $R(S)$ that are not reducible to its descriptive means.

Such an entity registers in $R(S)$: it is present there, interaction with it is possible, its properties admit measurement. But some aspects of its arrangement are not legible in the categories of $R(S)$.

Light serves as an empirical example. It is present in the $R(S)$ of our point, it registers, and it is measured. Some of its properties — in particular those taken in our $R(S)$ as zero or as boundary values — indicate that light does not fit completely within our $R(S)$.

This is a statement about the limit of $R(S)$: it contains something that goes beyond its descriptive means.

It follows that $R(S)$ is not closed relative to its own content. Elements may enter $R(S)$ whose arrangement is not derivable from its internal categories.

9. Summary

The geometry of reduction describes four levels of structural relation.

First — the relation between S and $R(S)$: the full structure and its reduced description, always separated by a constitutive difference.

Second — the division of the components of S into those symmetric and those asymmetric relative to R , which in $R(S)$ yields laws and residues respectively.

Third — the parametric nature of R , which generates a space of points of reduction within which a subset of points admits an observer.

Fourth — the double nesting of reductions at a point that admits an observer: the external reduction $S \rightarrow R(S)$, and the internal reduction that structures individual dimensions of $R(S)$ through the observer's memory.

Within $R(S)$, entities may be present that are not reducible to its descriptive means. This fact indicates the geometry's own boundary: $R(S)$ describes what fits within it and makes no claim about what does not.
